

Multiple Choice (10 marks)

1. Applications developed by programming languages like _____ and _____ have this common buffer-overflow error.
 - (a) C, Ruby
 - (b) Python, Ruby
 - (c) C, C++
 - (d) Tcl, C#
2. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
 - (a) Joined
 - (b) Authenticated
 - (c) Submitted
 - (d) Shared
3. _____ is the anticipation of unauthorized access or break to computers or data by means of wireless networks.
 - (a) Wireless access
 - (b) Wireless security
 - (c) Wired Security
 - (d) Wired device apps
4. Buffer-overflow may remain as a bug in apps if _____ are not done fully.
 - (a) boundary hacks
 - (b) memory checks
 - (c) boundary checks
 - (d) buffer checks
5. The _____ was a huge marketplace of Dark Web specifically famous for selling of illegal drugs & narcotics as well as you can find a wide range of other goods for sale.
 - (a) Silk Road
 - (b) Cotton Road
 - (c) Dark Road
 - (d) Drug Road

True / False (10 marks)

Answer True or False

6. True or False: The AH Protocol guarantees both data integrity and privacy for the information being transmitted.
7. True or False: Buffer overflow vulnerabilities are primarily caused by insufficient processing power during application execution.
8. True or False: The Heartbleed bug allows an attacker to perform a read outside the bounds of a buffer, exposing sensitive data in memory.
9. True or False: A buffer overflow occurs when a pointer is used to access memory not allocated to it.
10. True or False: Encapsulating Security Payload (ESP) is a protocol that belongs to the Internet Protocol Security (IPsec) suite.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the number of subsequences having product smaller than k for the given non negative array.
12. Write a function to remove an empty tuple from a list of tuples.

End of Paper